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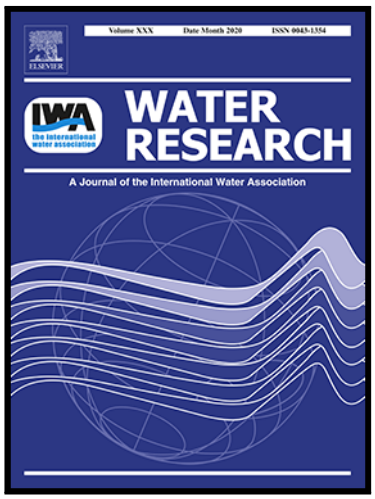
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Reconstruction of corrupted datasets from ammonium-ISE sensors at WRRFs through merging with daily composite samples

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Highlights

- Recalibration of A-ISE sensors with single grab samples is highly problematic.
- Useful information can be extracted from otherwise problematic, corrupted data.
- Merging A-ISE observations with composite samples increases data reliability.
- The corrected data gives sharper parameter distributions for a software sensor.
- Eight to sixteen weeks of data are sufficient for training a good software sensor.

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Title: Reconstruction of corrupted datasets from ammonium-ISE sensors at WRRFs through merging with daily composite samples

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Keywords:

Ammonium; ion-selective electrodes; data merging; software sensor; sensor maintenance

Abstract:

Long-term, continuous datasets of high quality are important for instrumentation, control, and automation efforts of wastewater resources recovery facility (WRRFs). This study presents a methodology to increase the reliability of measurements from ammonium ion-selective electrodes (ISEs). This is done by correcting corrupted ISE data with a data source that often is available at WRRFs (volume-proportional composite samples). A yearlong measurement campaign showed that the existing standard protocols for sensor maintenance might still create corrupted dataset, with poor sensor recalibrations responsible for abrupt and unrealistic jumps in the measurements. The proposed automatic correction methodology removes both recalibration jumps and signal drift by using information from composite samples that already are taken for reporting to legal authorities. Results

showed that the developed methodology provided a continuous, high-quality time series without the major data quality issues of the original signal. In fact, the signal was improved for 87% of days when a reference sample was available. The effect of correcting the data before use in a data-driven software sensor was also investigated. The corrected dataset led to noticeably smaller day-to-day variations in estimated NH_4^+ loads, and to large improvements on both median estimates and prediction bounds. The long time series allowed for an investigation of how much training data that is required to fit a software sensor, which provides estimates that are representative for the entire study period. The results showed that 8 weeks of data allowed for a good median estimate, while 16 weeks are required for obtaining good 80% prediction bounds. Overall, the proposed method can increase the applicability of relatively cheaper ISE sensors for ICA application within WRRFs.

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1. Introduction

The wastewater sector is moving towards more digitalization, and more instrumentation, control and automation in an effort to improve infrastructure planning and operational performance (Olsson et al., 2014; Yuan et al., 2019). Nitrogen removal is among the major processes driving the design and affecting the operations of Water Resources Recovery Facilities (WRRFs), and thus one of the key components that are targeted by these efforts. Several WRRF control strategies imply monitoring of nitrogen forms, mainly ammonium, NH_4^+, and nitrate, NO_3^-, for defining set points for e.g. aeration or recycling rates during removal processes (see the examples in Olsson et al. (2014); Sweeney and Kabouris (2018); Yuan et al. (2019)), or to limit the emission of greenhouse gasses like N_2O (Andalib et al., 2018; Ekström et al., 2017). Furthermore, information on incoming nitrogen concentrations and loads can also be used by optimization approaches based on Model Predictive Control (e.g. Stentoft et al., 2019; Vrečko et al., 2011).

Long-term, continuous time series of relevant variables in the system are key for supporting all of these efforts. For WRRFs, these time series can e.g. be used for the development of influent generators (Martin and Vanrolleghem, 2014), to support the development of plant wide models (Cierkens et al., 2012), and to set up software sensors and/or data-driven models (Newhart et al., 2019). Online sensors that measure NH_4^+ and NO_3^- are either based on ex-situ (analysers) or in-situ installations (Béltran et al., 2015). While the first analyse a pre-filtered wastewater sample, which is pumped out of the monitored stream, the latter are placed directly in the wastewater stream. Generally, ex-situ commercial sensors utilize colorimetric or ion-selective electrodes (ISE), while in-situ applications are based on ISE or UV-sensors (Gernaey et al., 1998). The two approaches obviously differ in terms of accuracy and precision, reliability and maintenance requirements, and thereby also in their financial costs. Ex-situ sensors based on colorimetric techniques are often judged as more reliable, but require important financial

investments. Conversely, in-situ ISE sensors have the advantage that they are significantly cheaper, have very little lag time in their data delivery, and do not require any chemicals for their analysis (Rieger et al., 2002).

Ammonium-ISE (A-ISE) sensors have been tested in multiple environments (Winkler et al., 2004): in harsh environments (sewer, WRRF) the A-ISE performance was strongly affected by clogging and fouling (requiring extensive maintenance and sensor cleaning), while low target-ion concentrations challenged measurements in cleaner streams (such as WRFF outlets). This was also observed by Papias et al. (2018) when using A-ISE for monitoring a wetland treating WRRF effluent. The main issue with measuring in low-concentration environments is that, other ions (especially K^+ and Na^+) can interfere with the ion-selective membrane (Cecconi et al., 2019; Winkler et al., 2004).

ISE sensors can also be strongly affected by drift, as experienced by Papias et al. (2018) after about 30 days of operation. Tendency to drift immediately after installation was also observed by Ohmura et al. (2019) when benchmarking multiple ISE sensors for pH measurements in a laboratory study. Signal degradation and sensor ageing are caused by (1) reversible fouling, which is more severe closer to the WRRF inlet and which can be cleaned with a piece of cloth, and (2) irreversible fouling, which seems to be independent of the location in the plant and which cannot be cleaned (Cecconi et al., 2019).

Appropriate Standard Operating Procedures (SOP) for sensor maintenance are thus essential to ensure good quality of the collected data, while improper SOP can irremediably deteriorate the sensor signal. For example, Cecconi et al. (2019) showed how the use of non-representative wastewater samples to recalibrate A-ISE sensors can significantly worsen their performance. In this case, ex-situ reference samples (i.e. collecting wastewater in a bucket and using it for sensor calibration) were recommended.

Rieger et al. (2010) proposed a procedure for obtaining high-quality data, which requires performing additional intensive monitoring campaigns. Both of these approaches produce additional workloads and costs for the WRRF operators. Similarly, SOPs adopted by several scientific studies (Alferes et al. (2014), for example, suggested a weekly sensor maintenance) might be regarded as too demanding by WRRF operators, who need to navigate a trade-off between the relatively lower price tag of an A-ISE and the maintenance costs required to ensure high-quality measurements.

There are examples of methods for retrieving useful information from unmaintained or poorly maintained sensors (e.g. De Mulder et al., 2018; Schneider et al., 2019). Additional data validation methodologies have also been described by several studies (Alferes et al., 2013; Leigh et al., 2019), with some specifically considering sensor drift (Haimi et al., 2016; Rieger et al., 2005; Thomann et al., 2002), data reconciliation and gap filling (De Mulder et al., 2018), and online estimation of measurement uncertainty (Kahiluoto et al., 2019; Näykki et al., 2015). Papias et al. (2018) attempted a drift correction with a simple linear relationship over time, while Schneider et al. (2019) extracted useful information from the signals of unmaintained sensors. Most of these methods are, however, semi-automatic, i.e. they still require some degree of intervention from a data scientist.

This study introduces an automatic data correction methodology for A-ISE sensors, which relies on reference measurements that are commonly available at many WRRFs. The proposed method increases the reliability of A-ISE data and extracts information from otherwise corrupted time series without establishing new, resource-demanding SOPs for WRRF operators. Data from a 1-year measurement campaign performed by the largest wastewater utility company in Denmark at a WRRF in Copenhagen were used. Sensor maintenance was performed by closely following the SOP provided by the sensor manufacturer, and the collected measurements thus represent a “state-of-the-art” dataset for everyday,

in-field sensor operations. Nevertheless, sensor drift and recalibration jumps make the raw data difficult to use in practice. In this study, we therefore:

1. Develop an automatic method for correcting the corrupted, raw time series by removing the effects of improper sensor recalibration and by reconstructing a more reliable signal. The developed method utilizes existing, commonly available, composite samples as reference, and it thus reduces the need for additional maintenance resources;
2. Evaluate the performance of the automatic correction method when applied in online WRRF operations as well as in an offline context (correction of historical data);
3. Investigate the impact of the corrected data for operating a WRRF (analysis of daily NH_4^+ profiles) and for the application of a data-driven, online software sensor;
4. Discuss the need for modifying existing SOP and maintenance procedures for A-ISE sensors providing a trade-off between the need to maintain high-quality data while reducing the risk of overzealous maintenance.

2. Materials and methods

2.1. Case study: Damhusaaen WRRF

The Damhusaaen WRRF is operated by BIOFOS, the largest Danish wastewaster utility, and is one of three major plants in the city of Copenhagen, Denmark. The plant receives wastewater from a 55 km^2 combined sewer catchment, consisting of a mix of households and industries. The plant has a capacity of 350,000 PE and is an alternating plant (Isaacs and Thornberg, 1998) with biological nitrogen and phosphorous removal as well as chemical phosphorous precipitation. Figure 1 shows an overview of the plant layout, the main hydraulic pathways, and the location of the monitoring locations used for this study. The wastewater stream is sampled after primary treatment (location 1), just before a pumping station. An A-ISE sensor is installed downstream of the pumping station, in the inlet channel to the

biological treatment tanks (location 2). The placement of the sensor has been defined by BIOFOS staff for logistical reasons, i.e. to provide easy access and to limit fouling caused by fat and grease, which are known factors affecting the reliability of both in-situ and ex-situ sensors (Rieger et al., 2002). Data for this study were collected for almost a full year in the period of June 4th 2018 to May 15th 2019.

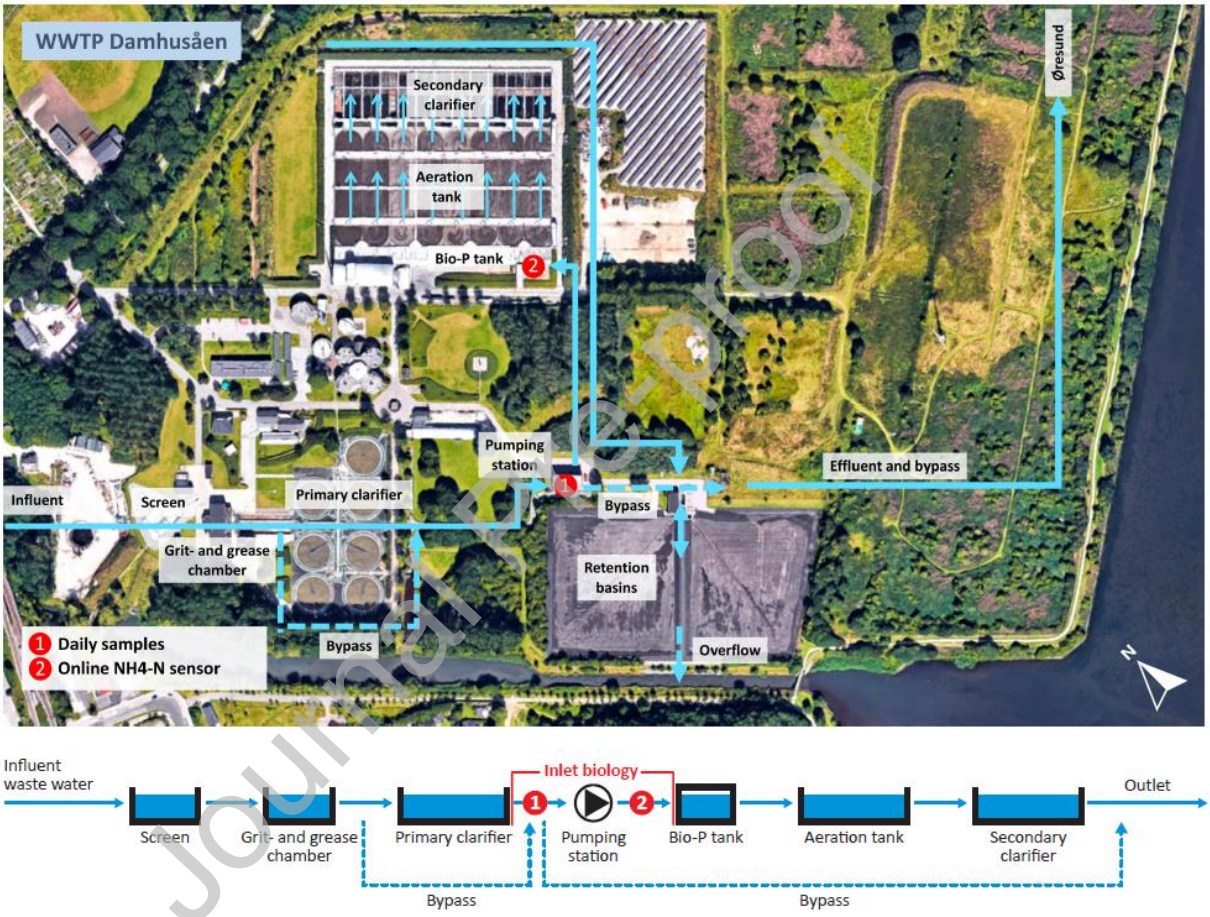


Figure 1: Aerial photo of the WRRF with main hydraulic pathways indicated (top) and a schematic layout (bottom). The red points designate measurement locations.

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2.2. Available data

2.2.1. Flow observations

Flow observations are collected after the central pumping station by using a magnetic inductive flow meter with a one-minute resolution. Basic data quality control (DQC) is automatically performed by the data collection platform, checking for out-of-sensor-range, sensor failures (flat lines), and sudden shifts. Small gaps in the time series of less than 10 minutes are filled by linear interpolation. Since the pump's operation strongly affect the measured flow data, a moving average with a centred window of two hours is used to remove the spikes caused by pump operations from the influent signal. Flow data are available for the full period of study.

2.2.2. Volume-proportional composite samples

The Danish legislation requires WRRF operators to report on the plant's yearly performance to the Danish Environmental Protection Agency (DK-EPA). The performance is evaluated by taking 24-hour volume-proportional composite samples (cf. definition in Ort et al. (2010)) at the inlet, outlet, and (when happening) the bypass of the plant. The required sample frequency depends on the plant capacity: for Damhusaaen, the DK-EPA requires at least 12 annual samples of the influent and 24 samples of the effluent. The plant operators take additional samples for internal analysis to obtain a better overview of the plant performance, with a total of 48 samples being collected annually at location 1 (Figure 1). The auto-sampler is not started on a fixed schedule as the operators intend to collect representative samples of all days of the week over a year. The sampling is thus spread out over the year, and there is approximately one sample per week on average with typical intervals between sample ranging from 4 to 10 days. The auto-sampler that was used for this study is a MAXX SP Zone 2 auto-sampler with a total sample capacity of 9 litres per sampling day. The auto-sampler collects wastewater aliquots of 60 ml, resulting in a maximum of 150 samples per day. The sampling volume interval is modified by the plant

operators depending on the expected inlet flow (i.e. if wet weather is expected during the sampling period). Unfortunately, the auto-sampler settings have not been documented, but the operators often utilize a value of 500 m^3 of wastewater passing by the sampler before a new aliquot is taken. A sampling day runs from 7AM to 7AM the following day. During heavy rainfall events, it is possible that the total capacity of the auto-sampler is exceeded and that the sampling thus ends prematurely. During the study period, 45 daily composite samples were collected. Three samples were discarded due to unrealistically high or low values after consultations with the operators, while four samples were discarded due to missing flow or A-ISE data for the same day.

2.2.3. Ammonium-ISE data

An ISE consists of an indicator electrode and a reference electrode. The reference electrode sits in a reference solution and has a constant electrical potential that is known to the manufacturer, while the potential of the indicator electrode will vary depending on the concentration of the target ion in the wastewater. The sensor measures the difference in potential between the two electrodes with a voltmeter. Knowing the potential in the reference solution and the measured difference between the reference and the indicator, one can estimate the concentration of a target ion in the wastewater through the Nernst equation:

$$E = E^0 + \frac{RT}{zF}\log(A) \quad (1)$$

Where E [V] is the electrode potential, E^0 [V] is a constant that is specific to the indicator-reference electrode pair, R is the gas constant [J $K^{-1}mol^{-1}$], T [K] is the temperature, z [-] is the charge of an ion, F is the Faraday constant [C mol^{-1}], and A [-] is the activity of the target ion. As there are different ions in raw wastewater, an "ion-selective" membrane only allows a specific ion such as NH_4^+ to reversibly bind to the indicator electrode. However, no membrane is 100% selective towards its target ion, and other

similar ions can cause interference. Accounting for this in equation 1, leads to equation 2 that is also called the Nikolskii-Eisenmann equation:

$$E_{meas} = c + \frac{RT}{z_i F}\log\left(A_i + \sum_{j=1}^{J} k_{ij} A_j^{\frac{z_i}{z_j}}\right) \quad (2)$$

Where E_{meas} is the measured potential difference, c is a constant, while subscript i and j refers to the target and the interfering ion, respectively. J is the total number of interfering ions, while k_{ij} is the selectivity coefficient and accounts for how well the membrane distinguishes between the target and the interfering ion. Equation 2, also known as the calibration curve of an ISE sensor, is a linear relationship between the measured potential difference and the logarithm of the activity, with an offset (c) and a slope parameter (RT/z_iF). Linearity only holds within a certain range of concentrations, and interference from other ions becomes an issue at very low concentrations of the target ion. At high concentrations, the assumption of concentration roughly equalling activity is not respected due to high ionic strength in the wastewater.

ISE sensors are known to drift over time with several identified causes: the reference potential can be affected by diffusion of molecules in the reference solution, the membrane can experience clogging, and there might be aging issues with various components in the sensor. The degree of drifting is dependent the type of sensor and the way it is used. The standard practice to minimize drift is to recalibrate the sensor regularly by re-estimating the offset and slope parameters in Equation 2. This is normally done by comparing sensor readings against laboratory analysis of grab samples. If just one grab sample is taken for comparison with sensor readings, it is called a one-point calibration, which only makes it possible to re-estimate one of the parameters in Equation 2. If both parameters are to be estimated a two- or multi-point calibration is needed.

The ISE sensor placed at location 2 at Damhusaaen WRRF is a HACH AISE sc probe, which provided data for the whole study period at two-minute resolution. The sensor corrects the signal for K^+, as this is the most important interfering ion in wastewater. Data were validated by using the same DQC procedure as for flow observations with additional visual checks to remove sensor maintenance periods, which are often not detected by automatic DQC.

The plant operators performed regular sensor maintenance, following a SOP that was agreed upon with the sensor manufacturer. The SOP required a manual cleaning of the sensor once a week with a wet piece of cloth. No chemical cleaning of the sensor head was performed during the study period. Grab samples were taken in the raw wastewater stream next to the sensor and a reading of the sensor observation at the same time was logged. According to the SOP, the sensor should be recalibrated if deviation between the grab sample and the sensor reading is greater than 10%. The utility company chose the value of 10% based on their standard internal practice, which is also applied to other sensors across their plants. Recalibration was done by following the sensor manufacturer's recommendations, suggesting a one-point calibration, where only the offset parameter is updated. In practice, the process of recalibration consisted of the operator typing the result of the laboratory analysis of the grab samples into a HACH SC1000 controller that directly connects to the ISE sensor. Proprietary software inside the controller then automatically updated the offset parameter of the calibration curve. Overall, the collected data are thus representative for a typical everyday use of such sensor by scrupulous operators in daily WRRF operations.

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2.3. Data correction method

The proposed data correction method consists of three steps aiming at removing the large discontinuities due to sensor recalibration as well as any sensor drift by using information from 24hr volume-proportional samples. The signal reconstruction aims to preserve the information on relative intra-daily variations from the ISE signal, while adjusting the absolute values to a more reliable estimate. The use of a 24-hr sample implies that the correction is not affected by inter-daily variations (e.g. weekdays vs. weekends), as long as the sensor data and the composite samples have been collected over the same period. This correction also removes the signal disruption caused by sensor recalibration based on non-representative grab samples. The entire process is illustrated in Figure 2.

1. **Remove discontinuous recalibration "jumps":** The first step identifies and removes discontinuities that are caused by manual sensor recalibrations in a measured time series, providing a smooth series of observations. Jumps were identified thanks to the operators' maintenance logbook, where recalibration events were recorded. A recalibration event causes an instantaneous jump in the measured series as the one-point calibration procedure adjusts the offset parameter in Equation 2. A change to the offset parameter does not only move the daily average of the observations but also changes the magnitude of the intra-daily variations. Since the sensor calibration curve relates the measured E signal linearly to the logarithm of the NH_4^+ concentration (see Equation 2), it is necessary to operate on log-transformed data. The first step back-calculates erroneous jumps by using the difference between the mean of the 10 observations before the jump and the mean of the 10 observations after the jump (Figure 2a). This difference is either added to or subtracted from the observations after the jump, aligning the observations on both sides of the recalibration event (Figure 2b). This is done sequentially for all recalibration events in the considered period, generating a times series where no sensor recalibrations are visible. Because the jumps are removed by evaluating the average concentration immediately before

and after a jump, it is important to choose a window of time for averaging that is long enough to smooth out random noise in the data. The time window should also not be so long that the dynamics in the system have led to a real change in concentration, e.g. during the ramping of the daily peak. For this study, a window size of 10 observations provided a good compromise between the two aspects.

2. **Emulate composite sampling:** This step calculates a 24-hour average concentration based on ISE data, emulating the auto-sampler's volume-proportional sampling of the actual wastewater stream. First, the cumulative wastewater volume during the sampling day is calculated (Figure 2c). According to the typical settings at Damhusaaen WRRF, samples are taken every 500m^3 of passing water volume. The measured NH_4^+ concentrations at corresponding sampling times are then extracted from the time series (corrected at step 1), and their average is calculated, creating a composite sample from the sensor data (Figure 2d). On rainy days with large incoming wastewater volumes, the auto-sampler might run out of its 150-sample capacity. This is emulated by stopping the sampling from the sensor data at the 150th sample as well.

3. **Adjust ISE series with composite values:** This step adjusts the ISE values to fit the measured composite samples. As in step 1, the adjustment is performed on log-transformed data. First, a correction factor is determined for the days when composite samples were taken as the ratio between measured and emulated composite values (Figure 2e). Linear weighting between these correction factors are then used to estimate correction factors for days between two composite samples. Finally, the corrected time series is obtained by multiplying the correction factors to the data generated at step 1 (Figure 2f).

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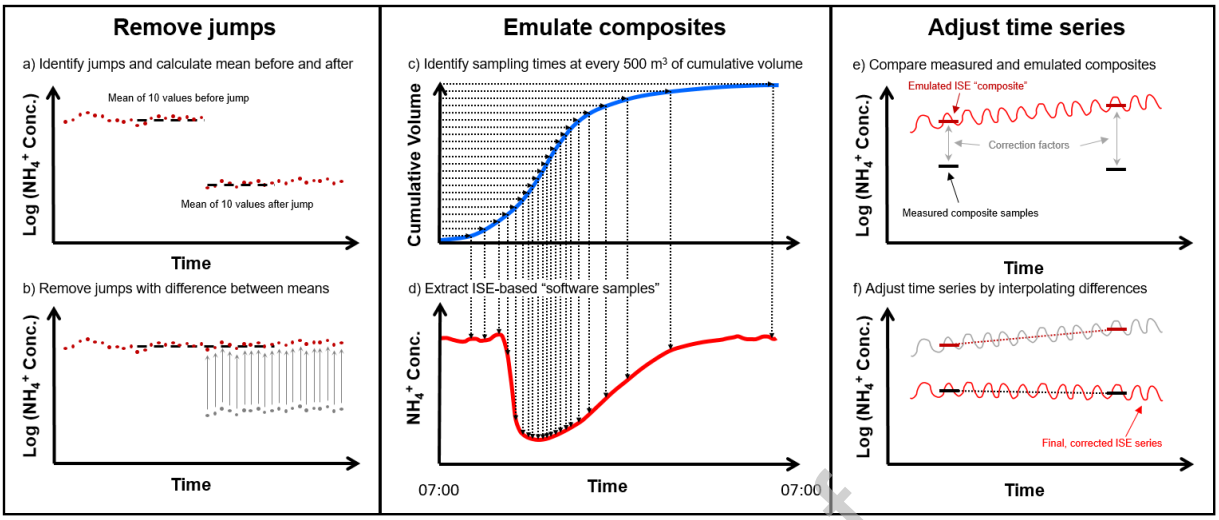


Figure 2: Sketch of the different steps in the data correction method. It is worth noting that the time scales are different for each column. On the left, the time axis is a couple of hours, in the middle it is 24 hours, while it is on the order of two weeks in the right column.

2.4. Offline vs online applications

The proposed correction method can be applied in offline, desktop settings (i.e. to correct data collected during past monitoring campaigns) and in online, operational settings (i.e. measurements are corrected as soon as they are recorded by the sensor). For offline applications on historical data, the correction of a given ISE data point can be based on its closest past and future composite samples. For online applications, the correction procedure can only be based on past composites samples, since future samples are obviously not available. In this case, the correction factors are kept constant and equal to the one estimated at the most recent composite sample day (i.e. there is no interpolation between two composite sample days).

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2.5. Evaluation of the corrected signal

The performance evaluation of the correction method depends on whether it is used in offline or online mode. The offline performance is evaluated with a "leave-one-out" analysis, where a single composite sample is removed from the dataset and the full correction procedure is performed with the remaining samples. The relative error ε_{raw} for the raw signal on the removed day is calculated as:

$$\varepsilon_{\mathrm{raw}} = \frac{C_{\mathrm{raw}} - C_{\mathrm{comp}}}{C_{\mathrm{comp}}} \quad (3)$$

Where C_{raw} [mg l^{-1}] is the concentration from the emulated composite sample (Step 2) based on the raw ISE signal, and C_{comp} [mg l^{-1}] is the concentration from the removed composite sample measured in the laboratory. The relative error ε_{corr} for the corrected data on the removed day is then calculated as:

$$\varepsilon_{\mathrm{corr}} = \frac{C_{\mathrm{corr}} - C_{\mathrm{comp}}}{C_{\mathrm{comp}}} \quad (4)$$

Where C_{corr} [mg l^{-1}] is the concentration from the emulated composite sample (Step 2) based on the corrected ISE signal. The correction procedure has improved the value of the ISE data if $\varepsilon_{\mathrm{corr}}$ is smaller than $\varepsilon_{\mathrm{raw}}$. This procedure is then repeated for all composites in the measurement campaign.

The performance of online applications of the method is done for each composite sample at a time. Here, the correction factors that were calculated for the most recent composite sample are extrapolated forward to the ISE data for the day of interest. Equation 4 is then applied to the online corrected signal.

2.6. Application to a data-driven software sensor

The benefits of using the correction method in combination with a data-driven software sensor were also evaluated. The modelling setup for the software sensor was based on the model initially presented by Vezzaro et al. (2020), where it was used for forecasting flow and NH_4^+ at the inlet of a WRRF. This

model simulates incoming NH_4^+ daily loads at the inlet of plant (F_{NH_4}) by using a widely applied simple second-order Fourier series (e.g. Bechmann et al., 1999):

$$F_{NH_4}(t) = \mu + \sum_{k=1}^{2}\left(\alpha_k sin(2\pi kt) + \beta_k cos(2\pi kt)\right) \quad (6)$$

where μ corresponds to the daily mean of the NH_4^+ load [kg/hr], and α_k and β_k are Fourier coefficients that reproduce the diurnal pattern. Since the ISE sensor is located after the primary clarifiers, their volume will cause the diurnal pattern to be attenuated and delayed before it reaches the sensor location. Equation 6 thus requires a modification, where the inflow signal is filtered by using a series of continuously stirred tank reactors with a total volume equivalent to the total clarifiers' volume. The number of tanks in series (three in this case) were defined according to the attenuation observed in the measurements. The NH_4^+ concentration (C_{NH_4}) is then calculated as:

$$C_{NH_4}(t) = \frac{F_{NH_4}(t)}{Q(t)} \quad (7)$$

Where $Q(t)$ is the measured inlet flow.

The software sensor is trained by estimating the parameters of Equation 6 for each day within a user-specified time window. The estimation procedure consists of minimizing the root mean squared error between simulated and observed data with a standard Nelder-Mead algorithm (R Core Team, 2019). This provides an ensemble of daily profiles that contains as many members as there are days within the time window. The ensemble allows for considering inter-daily variations (which are often neglected by models only considering a single typical profile) and it is used for forecasting or for filling information gaps in periods where NH_4^+ concentrations and loads are missing, thus allowing the WRRF operations to swiftly continue.

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To evaluate the influence of the amount of training data for estimating the software sensor, ensembles of different sizes are compared. Only weekdays' profiles are considered (i.e. no weekends). The ensembles generated by using different time windows (ranging from 2 to 16 weeks, and sweeping across the whole dataset) are compared to the results obtained by using all the 130 dry weekdays.

The performance of the software sensor is investigated with the following indicators:

(1) Mean Absolute Relative Error (MARE) of the ensemble median against the observations;

(2) Observation coverage of the 80% confidence interval as defined with the 10th and 90th percentile as lower and upper boundary;

(3) Interval Score (IS) as given by Equation 8 for the $(1-\alpha)\cdot 100\%$ confidence interval.

$$\mathrm{IS} = \sum \begin{cases} u-l & \text{for} \quad l < x < u \\ u-l+\dfrac{2}{\alpha}\cdot(l-x) & \text{for} \quad x < l \\ u-l+\dfrac{2}{\alpha}\cdot(x-u) & \text{for} \quad x > u \end{cases} \tag{8}$$

Where x is the observation at a given point in time, l is the lower boundary of the confidence interval, u is the upper boundary of the confidence interval, while α defines the confidence interval (e.g. α equals 0.05 and 0.2 gives a 95% and 80% confidence interval, respectively).

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3. Results and discussion

3.1. Assessment of signal quality

Figure 3a shows all the available inlet flow data. The average flow was around 3,000 m^3/hr during dry days with discernible seasonal variations: the dry weather flow was larger during winter months and in a

two-week rainy period in March 2019. These variations were due to groundwater infiltration-inflow and followed the seasonal variations in the groundwater table. Wet weather occurred when the measured flow exceeded 5,000 m^3/hr with stormwater peaks reaching 20,000-23,000 m^3/hr.

Figure 3b shows the raw NH_4^+ signal from the ISE sensor where a change in colour indicates a manual sensor recalibration event. The calibration time stamps are also shown in Figure 3c along with sensor cleaning dates and change of the sensor cartridge (November 2018). An indication of a "calibration" time means that the sensor was both cleaned and calibrated. Both infiltrated groundwater and stormwater dilute wastewater, leading to lowered NH_4^+ concentrations: these effects can be seen in the gradual changes over long time intervals (weeks, months - indicating groundwater infiltration) and in the sudden (hours, days) drops in the NH_4^+ concentrations due to stormwater dilution. In the period from September through November 2018, the sensor was maintained according to the original SOP with frequent cleaning and comparison of the deviation between ISE values and grab samples (approximately once a week). There were several exceedances of the 10% deviation threshold in this period, which led to frequent recalibration of the sensor. Two of these were in hindsight due to grab samples with unrealistically high concentrations, which should not have been used for recalibration in the first place. This maintenance procedure changed from early December 2018 to late February 2019 when the sensor was left unmaintained without any cleaning and recalibration. These changes in maintenance schedules were not actively planned as part of this study, but they provide a realistic picture of sensor operations outside an academic setup (including effects of operators' vacations, sick days, etc.)

Figure 3d shows the raw sensor signal (red) and the corrected time series (purple) for the whole dataset. The measured concentrations from the 24-hour volume proportional samples are also shown (black squares). Figure 4 shows an enlargement of a period with intensive maintenance (Figure 4a), and a

period with apparent sensor drift (July 2018 - Figure 4b). During the period with intensive sensor maintenance, the signal showed large jumps in concentration of up to 30-40 mg/L before and after a sensor re-calibration. After these jumps, the signal sometimes reached physically unreasonable values above 90 mg/L. This raises serious concerns on the representativity of the raw ISE signal. If this was the only available data source, it would be very difficult to determine which parts of the dataset that can be regarded as reliable. Figure 4a shows how the corrected signal provides a continuous, more realistic time series without the large disruptive jumps. While the dry weather concentrations varied between 30 to 90 mg/L in the raw series, they consistently stayed around 50 mg/L in the corrected signal. From a qualitative perspective, the corrected values seem much more reasonable as they do not show unrealistic changes in the dry weather concentrations.

The long period without sensor maintenance (Figure 3b) shows a downward trend in the raw ISE values over time until the next sensor recalibration (late February), when the signal jumps back up. A similar situation was observed in July 2018 (Figure 4b), where a period without maintenance resulted in an increasing trend in NH_4^+ values over time. However, the analysis of the ISE signal alone does not allow identifying if these trends are caused by sensor drift, variations in groundwater infiltration that dilutes wastewater (as seen in the flow observations from January 2018), changes in the wastewater influent loads (due to e.g. summer holidays in July), or a combination of these. The additional information provided by the composite samples thus enables a more thorough analysis, which, for example, shows how the ISE signal drifted away from measurements during these two periods. The correction method provides an adjusted signal that does not drift over time while still preserving the natural intra-daily variations in NH_4^+ concentration (Figure 4b).

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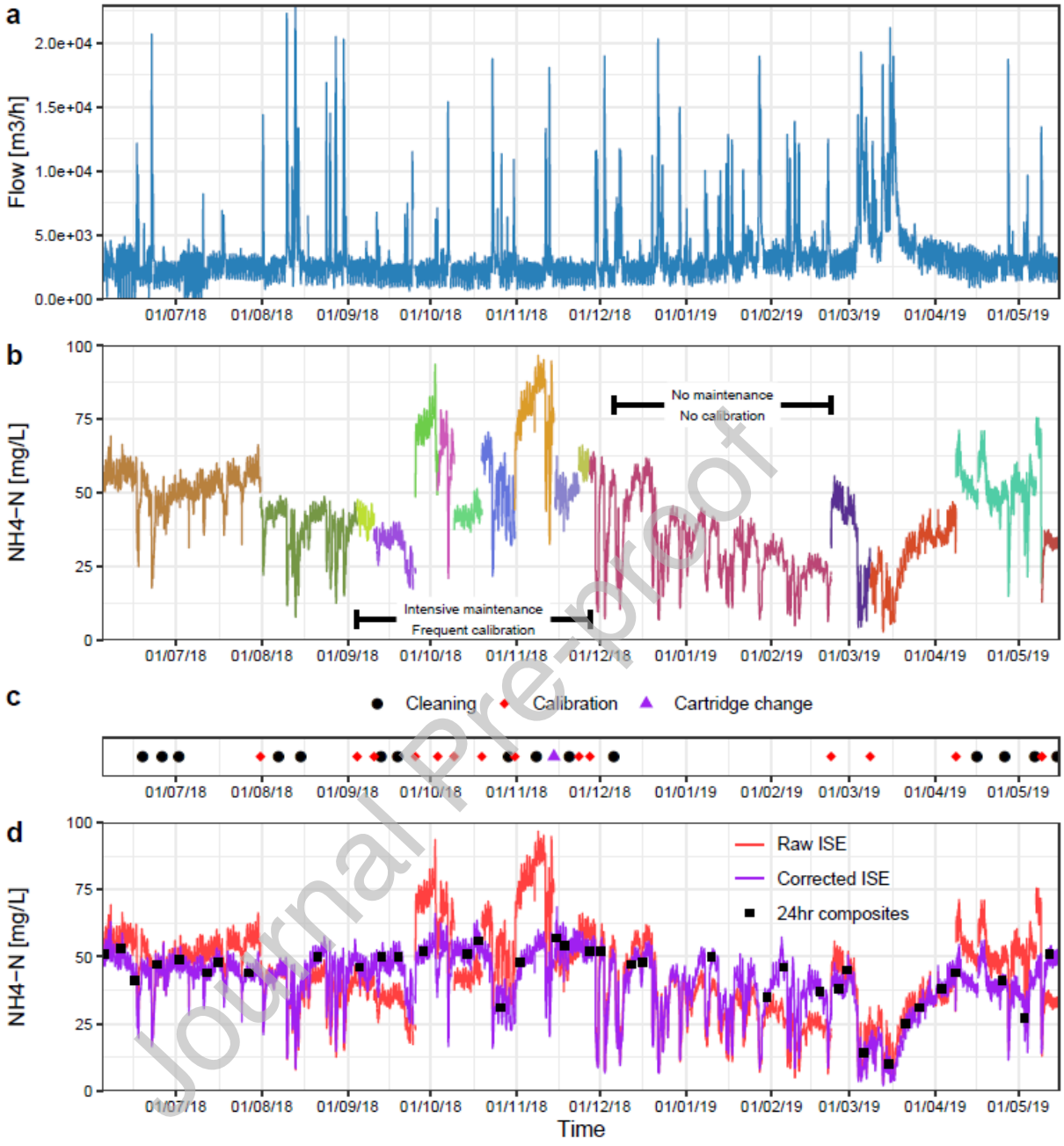


Figure 3: (a) Flow data, (b) raw ISE data with each calibration period in a separate colour, (c) indication of various types of sensor maintenance, (d) raw vs. corrected ISE signal vs. 24hr composites.

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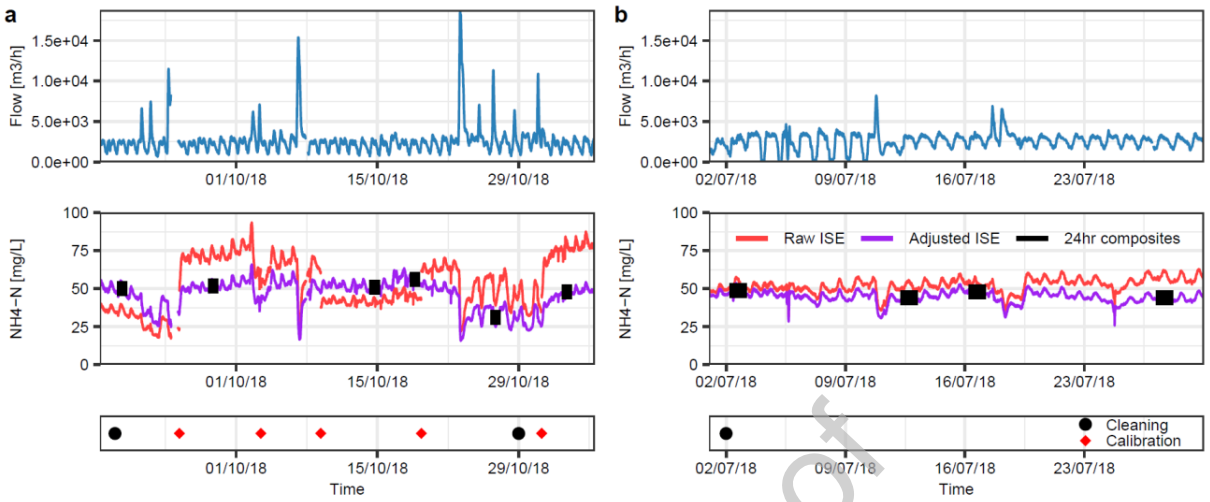


Figure 4: Zoom of part of the period with frequent maintenance (a, left column), and zoom of a period with sensor drift (b, right column).

Figure 5 compares the quantitative performance of the raw ISE signal against the two application modes (offline and online). Figure 5a shows the relative error between the measured composite samples and their ISE-based equivalents for the raw and corrected signals, as calculated by Equation 3-5. Here, it is seen that the offline correction produces smaller errors (quartile range of -8% to +7%) than the raw signal (quartile range of -13% to 24%). The online correction is, as expected, worse than the offline setup (quartile range of -15% to 12%) but it still provides an improvement compared to the raw signal. While the boxplots (Figure 5a) summarize the overall performance over the entire measurement campaign, Figure 5b,c show the errors for each individual sampling day. A red point indicates that the raw signal outperforms the corrected signal for a given day, while a purple or blue point signifies that the corrected signal is superior. Figure 5b,c show that there are a small fraction of days where the raw signal is best: overall, the offline correction method outperforms the raw signal in 87% of the 38 sampling days, while the online correction is superior in 71% of the days.

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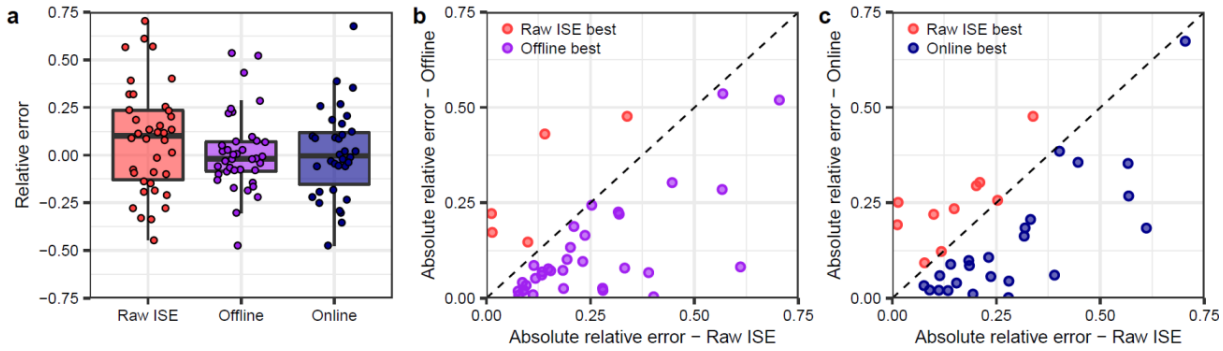


Figure 5: Comparison of measurement errors for (a) all three methods over the entire period, (b) raw vs. offline correction, (c) raw signal vs. online correction. Each point on the graphs is a day where composite samples were taken. Notice that (a) displays the relative error, while (b,c) shows the absolute relative error.

3.2. Software sensor performance

3.2.1. Distribution of daily profiles

Figure 6a,b shows the daily pattern of NH_4^+ loads for all 130 dry weekdays in the measurement campaign, with line colours corresponding to the calibration periods shown in Figure 3b. Figure 6a shows the daily load based on the raw data, where the classical daily wastewater profile is recognisable. Figure 6b provides the same information, but based on the offline corrected dataset. The thick black lines on the two graphs show the 10th, 50th, and 90th percentiles for all analysed days. The 50th percentile (i.e. the median) shows the measured load on a typical day, while the profiles outside the 10% and 90% bounds only represent 13 days (i.e. 10% of the profiles). A comparison between Figure 6a and 6b clearly shows that the day-to-day variability of daily loads is significantly reduced. For practical modelling purposes, this reduction will reduce the uncertainty in the estimation of daily loads. In fact, using the raw data would result in a large overestimation of both intra- and inter-daily variations.

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Figure 6c,d show the results of parameter estimations for the data-driven software sensor (section 2.6) in terms of full parameter distributions, while Table 1 summarize the median estimates and the range between the 10% and 90% quantiles. Parameters are estimated for each individual dry weekday in the campaign and summarised by their empirical distributions. By comparing the two graphs, it is seen that there is a large difference in the spread of the μ-parameter between the raw and corrected data (Table 1 shows more than 50% reduction in the 10%-90% interquantile range). Since this parameter represents the daily NH_4^+ mean, this is a direct consequence of the tightening in the ranges shown in Figure 6b. Table 1 also shows that the median of the parameter distributions are similar for the raw and corrected datasets, which indicates that there were no particular biases in the parameters estimated on the raw data. However, there are large reductions (25.8-34.5%) in the spread for three of the four shape parameters. The parameter distributions shown in Figure 6d can be used in stochastic models with the purpose of forecasting and to be used in e.g. Model Predictive Control (Stentoft et al., 2019).

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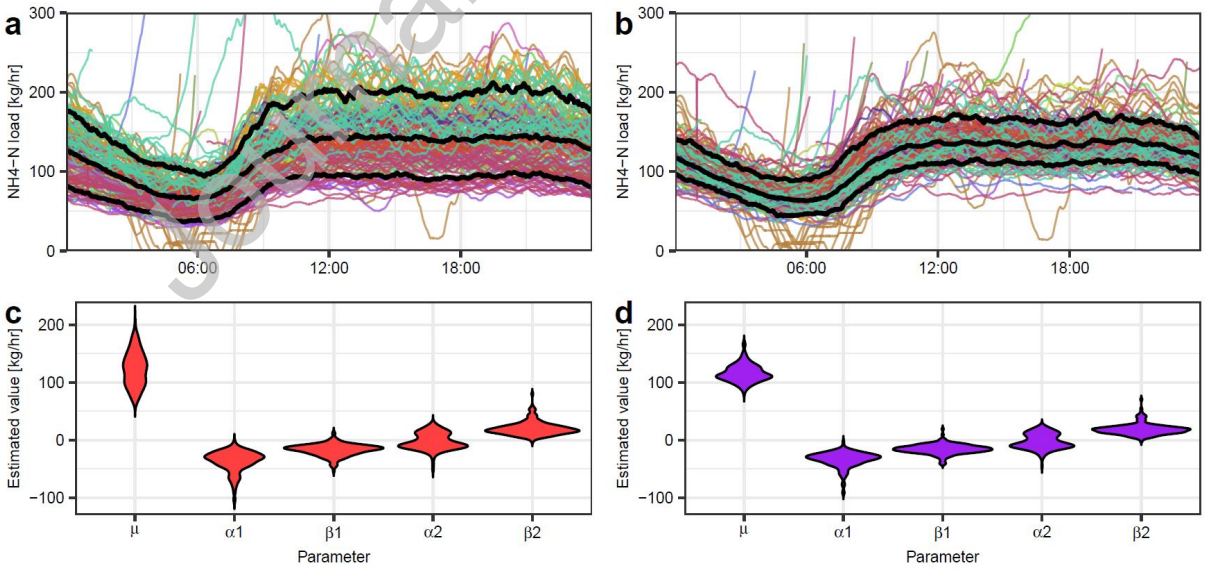


Figure 6: Daily profiles of NH_4^+ loads for 130 dry weekdays estimated with raw and corrected data, respectively (a, b). The colour of each line indicates which sensor calibration period from Figure 3 a given day belongs to. The thick black lines indicate the 10th,

50th, and 90th quantiles of the 130 daily profiles. Distributions of estimated parameter values in the software sensor for the raw and corrected signal, respectively (c, d).

Table 1: Median estimates and interquantile range (difference between 10% and 90% values) for the parameters of the software sensor.

	Median value			**Interquartile range**		
Parameter	**Raw [kg/hr]**	**Corrected [kg/hr]**	**Difference [%]**	**Raw [kg/hr]**	**Corrected [kg/hr]**	**Difference [%]**
μ	124.6	112.7	-9.5	77.3	34.4	-55.5
α1	-32.3	-31.3	*-3.0*	45.3	29.6	*-34.5*
α2	-15.6	-15.7	*0.7*	23.8	17.7	*-25.8*
β1	-5.88	-5.58	-5.0	32.8	32.1	-2.5
β2	17.4	17.6	1.6	26.2	19.1	-27.1

3.2.2. Effect on performance

Figure 7 shows a qualitative example of the difference between training the software sensor on data from the raw ISE signal and the corrected signal. The figure also shows how the sensor is affected by the amount of data it has been trained on (2 weeks versus 16 weeks). When Figure 7c,e is compared to Figure 7d,f it is clear that training the software sensor on the corrected dataset improves both the median estimate and the prediction bounds. With the raw data, the median estimate can be seriously biased at just two weeks of training (c versus d), while the prediction bounds are much wider when the sensor is trained on a longer period (e versus f). Although the median estimate is much more robust with the corrected data, there is still some structural issues at just two weeks of training as the daily peak concentration is estimated to arrive slightly later than in the observations (d). There is also a sharp second peak each day in the median estimate, which is not present in the observation. Training on 16 weeks leads to a median estimate that almost perfectly tracks the observations during dry weather (d). The 80% confidence interval is also larger for the 16-week case, which leads to a better coverage.

The software sensor has not been trained on wet weather and all of the software sensors (c,d,e,f) show poor performance during the rain event on November 29-30. The observations show a slower onset and recovery of the stormwater dilution effect than the model, a consequence of the model structural

limitations of these simple dilution-based models. To obtain better software sensing during wet weather situations, it will be necessary to implement additional elements in the model structure. Such additions can e.g. be a delayed onset of the dilution effect, a gradual restoration of concentrations at the end of an event, or a first flush effect (see Langeveld et al. (2017) for how to implement these additions).

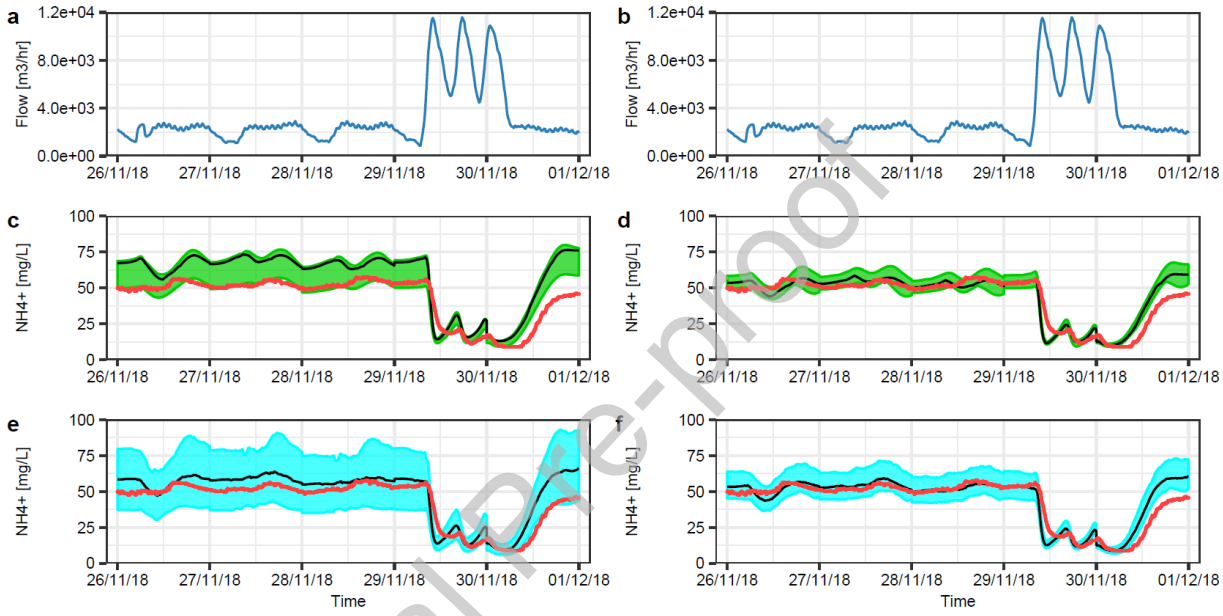


Figure 7: Illustration of software sensor performance when trained on the raw ISE data (left column) and the corrected data (right column). Observed flow (a,b), software sensor trained on two weeks (c,d), and software sensor trained on 16 weeks (e,f). The red dots are the observed NH_4^+ concentrations, the black line is the median software sensor estimate, and the coloured envelope is the 80% prediction bounds.

A further investigation of how the performance of the software sensor depends on the amount of data used for training is shown in Figure 8 (only shown for the corrected dataset). Figure 8a shows that the MARE of the ensemble median drastically improves with even small increases of the training time window. Already at eight weeks of training, the vast majority of the software sensors perform close to an ensemble that is trained on all available days (red line), with little visible difference when using a 16-

week window. Figure 8b,c shows that a larger training window is required to obtain good performance of the 80% confidence interval of the ensemble. While the ensemble median is quite stable already at eight weeks, the coverage can still be up to 25% lower than what is obtained by training on the full period. With two or four weeks of training, there is quite substantial variation in the performance depending on the exact period that is chosen for training within the study period.

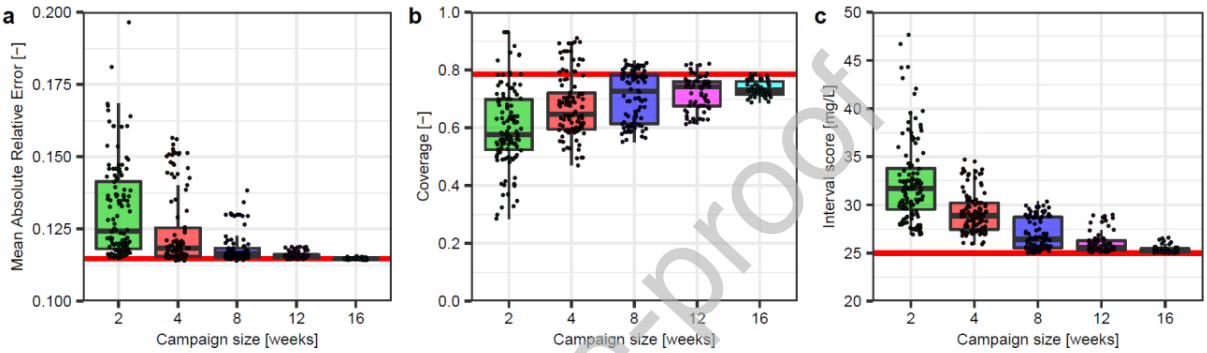


Figure 8: Performance of the software sensor as a function of the number of weeks that it is trained on. MARE of the ensemble median (a). Coverage (b) and IS (c) of the 80% confidence interval. The red line indicates the performance of a software sensor that has been trained on all available days. For a and c, lower values mean a better score, while the opposite is the case for b.

3.3. Implications for maintenance and operation of ISE sensors

Considering that the maintenance procedure was carried out by skilled operators of the largest Danish wastewater utility and with recurring consultations with the sensor manufacturer, we consider the data collected in this study as a good example of what a diligent operator can practically achieve in everyday WRRF operations. Through our conversations with the WRRF maintenance staff, it is clear that the erratic nature of the raw signal, with unrealistic jumps and low accuracy, seriously degrade trust in the value of ISE sensors. For recalibration of A-ISE sensors in raw wastewater, SOPs that rely solely on single, in-stream grab samples near the sensor are insufficient and should thus be abandoned. Our results also confirm the conclusion from Cecconi et al. (2019) that overzealous maintenance can potentially become

an issue for data quality, at least when it is based on single grab sample-based recalibration. Cecconi et al. (2019) concluded that good sensor recalibration can be achieved by comparing sensor readings against a grab sample from a bucket outside of the wastewater stream. However, this is considered impractical by BIOFOS operators, as it increases the maintenance efforts beyond what it is deemed reasonable for ISE sensors. Our correction method based on 24-hour composite samples removes these additional maintenance efforts, since the recalibration utilizes already existing measurements. In addition, the automatic procedure removes the risk related to human errors (e.g. “fat finger” errors when operating in the field). Thanks to the proposed automatic correction, the only maintenance that is necessary within the six-month lifetime of the sensor cartridge is physical cleaning of the sensor head. Based on our experience with operating a sensor behind the primary clarifier, we suggest that the sensor cleaning is kept at current frequency (e.g. weekly), while manual, in-field recalibration is done more sparingly and more thoroughly. Manual recalibration could reasonably be done with a two-point calibration that updates both offset and sensitivity parameters (e.g. monthly, or every other month).

Furthermore, using a 24hr composite sample follows the suggestion from Cecconi et al. (2019) to perform sensor recalibration against multiple grab samples taken over the course of a day instead of a single grab sample. Our results show how recalibration against composite samples improved the quality of the signal by reducing the uncertainty linked to the representativity of a single grab sample. It is worth noting that multiple single grab samples could be used to recalibrate both offset and sensitivity in eq. 2, while the mixed composite only allows for recalibrating one of the parameters.

Currently, the raw potentiometric signal (E) is not available to the operator, and the only available sensor output is the processed NH_4^+ signal. The raw potentiometric data, the parameters of the calibration curve, the selectivity coefficients for interfering ions, and other relevant settings are

considered intellectual property of the sensor manufacturer. If this information were available to the user, it would make it more straightforward to apply stochastic parameters that accounts for the uncertainty in various parts of the E-NH_4^+ transformation. Such a setup could provide uncertainty estimates on the ISE data that can be utilized for stochastic control routines (Stentoft et al., 2019).

3.4. Limitations, uncertainties, and future outlook

An essential assumption of the proposed method is the availability and the representativeness of the 24-hour composite samples. In a Danish context, these composite samples are required by law and the sampling protocols are defined by the Danish EPA. However, the method is applicable at any location where composite samples (also with sampling durations different from the 24 hours in this study) are routinely collected close to a sensor, and thereby making them available as reference values for the sensor correction.

Clearly, the performance of the correction method is sensitive to the sampling frequency of the reference samples. To ensure good performance, a minimum sampling frequency would be dependent on e.g. the rate of signal drift (which is also dependent on the specific brand of the sensor), and the composition of monitored wastewater. In this study, composite samples were available almost at a weekly frequency. Future research should investigate the effect of different sampling frequency on the performance of the correction method.

Since the composite samples are collected by following detailed protocols at Danish WRRFs, and they are used for legal requirements, their uncertainty is seldom questioned. However, as for any sampling operation, they are subject to equipment failures, sampling and lab uncertainties (Ort et al., 2010). Some of these uncertainties can be quantified, and they might be considered by a data correction setup

similar to ours (based on e.g. Kalman Filtering). Nonetheless, data quality control of the composite samples is a crucial step in the procedure. Furthermore, some locations (e.g. plant effluent) might significantly challenge the ISE sensors (due to e.g. too low concentrations, outside the ranges where there is a linear relationship between the measured electrical potential and the NH_4^+ concentration). This would clearly limit the accuracy and precision of ISE sensors, thereby hindering the recalibration procedure.

Whenever the auto-sampler and the ISE sensor are placed at different locations, a mathematical model is needed to align the two datasets (i.e. to estimate an equivalent of the composite samples at the sensor location or vice versa). Such a model will inherently have several uncertainties, which will be propagated to the correction method.

The proposed correction is essentially a one-point calibration of the offset parameter against a composite sample, and a two-point calibration of both offset and slope was not considered. Theoretically, the slope of the calibration curve varies with temperature and can be affected by membrane clogging. Also, offset and sensitivity might be affected by other components' ageing issues, while the importance of the temporal variability in the concentrations of interfering ions could also play a role (although a real-time K^+ compensation is performed by the sensor). Ohmura et al. (2019) showed that the sensitivity parameter was quite stable over time for pH-ISE sensors used in wastewater, suggesting that a one-point recalibration could be sufficient. However, this result was obtained in a controlled laboratory setup and with a pH sensor, and a different behaviour could be hypothesized in a field application at a WRRF. Because the proposed method only updates the offset parameter and only compares estimated daily averages from the ISE sensor and a reference sample, it is limited in its ability to ensure that intra-daily dynamics are correct. To check the sensor performance in representing these

dynamics, a two- or multipoint recalibration with comparison against sub-daily estimates would be needed. Since these data were not available in this study, a multi-point calibration was not investigated.

Further analysis should thus investigate the optimal frequency of sensor cleaning and composite-based recalibration, as well as the benefits of two- or multi-point recalibration. This would require the analysis of a fractionated, sub-daily reference sample, but since the infrastructure is already in place, this would just require a modification of the current monitoring protocols defined by the authorities.

The performance of any software sensor is dependent on the representativity of the training dataset. The software sensor investigated in this study assumes stationarity of the NH_4^+ loads generated in the catchment. Therefore, its performance might worsen in case of large intra-daily variations (e.g. weekend and tourism effects) that were not observed during the measurement campaigns. The software sensor evaluation suggests that a relatively short measurement campaign of 8-16 weeks is long enough to train a good software sensor for this case study. 8 weeks of training provided a good median estimate of the NH_4^+ concentration, while 16 weeks were required for a good estimation of the uncertainty bounds. This finding suggests that it might be possible to operate a sensor for a limited amount of time and still obtain good estimates of NH_4^+ for other periods. This finding is especially interesting for applications in more harsh measurement environments such as the plant inlet or upstream in the sewer system, where it is difficult to operate a sensor for long periods of time (e.g. for predicting concentrations upstream of combined sewer overflows). Future research will investigate how well this finding holds up in these environments. Also, the combination of this method with automatic online estimation of measurement uncertainty, as in the approach investigated by Kahiluoto et al. (2019) and Näykki et al. (2015), could increase the value of the reconstructed dataset.

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4. Conclusion

This study demonstrated how the reliability of an online water quality sensor (here an A-ISE for measuring NH_4^+ concentrations) can be improved when the raw signal, affected by several sources of interference, is combined with information from traditional sampling methods (here daily volume-proportional composite samples).

The proposed correction method reconstructed a more reliable signal starting from a corrupted data set and by using already existing information. The suggested approach removed the slow drifting of the sensor signal over time, as well as the effect of flawed manual calibrations. The analysis in this study showed that the suggested correction methodology improved the ISE estimate of NH_4^+ concentrations for 87% of the days where reference samples were available.

The time series of corrected NH_4^+ values contained small inter-daily variations and it was possible to fit a data-driven software sensor with reasonably tight parameter distributions. This makes the data more suitable for stochastic generation of inlet data for WWTP modelling as well as stochastic forecasting and control.

A quick investigation showed that there was good potential for setting up an ensemble-based software sensor that can fill gaps in periods where there is no data. The results showed that training a software sensor on the corrected rather than the raw data drastically improved both median estimates and prediction bounds. Training a software sensor on just eight weeks of data led to a good median estimate of NH_4^+ concentrations. 16 weeks of data were required for training in order to obtain good performance of the 80% confidence interval of the software sensor.

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Our results showed how existing SOPs for ISE sensors in raw wastewater should be modified and improved. It was found that even in a fairly protected measurement location (after the primary clarifier), and with qualified operators following a maintenance procedure in close collaboration with the sensor manufacturer, single grab sample adjustments were not sufficient for maintaining good signal quality. The resulting signal was very erratic during periods with frequent sensor maintenance and recalibration.

Our results support the hypothesis that the use of multiple grab sample for each sensor recalibration could provide more robust recalibration. Indeed, using composite samples for sensor data correction led to an improved signal. Based on the experience from this study, it is suggested that cleaning of the sensor head is performed frequently, e.g. once a week, while manual, in-field sensor recalibration is done more sparingly, e.g. once a month or every other month. In between those recalibration events, a reliable ISE signal can be obtained by using the proposed automatic correction procedure.

The proposed methodology can be applied to improve the reliability of online sensors for WRRF operations without requiring additional sampling efforts from the plant operators. It can also be applied to retrieve information from corrupted data of historical measurements campaigns. Overall, it will thus provide new possibilities for exploiting existing sensor data for better and more effective WRRF operations.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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